

Program 5

Design and implement C/C++ Program to obtain the Topological ordering of vertices in a given digraph.

```
#include<stdio.h>

void main()
{
    int a[10][10],t[10],indeg[10],n,SUM=0;
    int u,k=0,v;
    int i,j,stack[10],top=-1;
    printf("\n\n\t topological ordering \n\n");
    printf("enter the directed acyclic graph\n\n");
    printf("enter the no of vertex\t");
    scanf("%d",&n);
    printf("enter the adjacency matrix\n");

    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            scanf("%d",&a[i][j]);
        }
    }

    for(i=0;i<n;i++)
        indeg[i]=0;

    for(j=0;j<n;j++)
    {
        SUM=0;
        for(i=0;i<n;i++)
        {
            SUM+=a[i][j];
        }
        indeg[j]=SUM;
    }

    for(i=0;i<n;i++)
    {
        if(indeg[i]==0)
        {
            stack[++top]=i;
        }
    }

    while(top!=-1)
    {
        u=stack[top--];
        t[k++]=u;
        for(v=0;v<n;v++)
        {
            if(a[u][v]==1)
            {
                indeg[v]--;
            }
        }
    }
}
```

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                                if(indeg[v]==0)
                                {
                                    stack[++top]=v;
                                }
                            }
                    }

    printf("the topological sorting list\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",t[i]+1);
    }
}

```

Sample Input and Output:

topological ordering

enter the directed acyclic graph

enter the no of vertex 5

enter the adjacency matrix

0 1 0 1 0

0 0 0 1 1

1 0 0 1 1

0 0 0 0 0

0 0 0 0 0

the topological sorting list

3 1 2 5 4